MILITARY SPECIFICATION

CONTROL ASSEMBLY R.F. (UNIT 1) 11748205

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1. This specification covers the Control Assembly R.F. (Unit 1) used with the System Tester, Radar AN/VPM-2, 11748124.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Material, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-46357	Computer, Gun Direction, M18, Manufacturing Processes and Inspection of
MIL-S-48431	Solid State X-Band Source 10548182
MIL-P-48606	Power Supply Assembly (Unit 2) 11748133

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS (SEE 6.3)

US Army Armament Research and Development Command (ARRADCOM)

11748133	Power Supply Assembly (Unit 2): 11748133
11748168	Microwave Integrated Assembly #1
11748205	Control Assembly R.F. (Unit 1) 11748205
11748455	Microwave Integrated Assembly #5
11748460	Wire List An/VPM-2

PACKAGING DATA SHEET

11748205	Packaging of Control Assembly R.F. (Unit 1): 11748205

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item identification. The Control Assembly R.F. (Unit 1), hereinafter called the "Control Assembly," consists of microwave and digital components necessary to perform control and measurement functions required to accurately determine the radar operational status of the AN/VPS-2 Vulcan Radar System.

3.2 Design and construction. The Control Assembly shall be fabricated and assembled in accordance with Drawing 11748205, parts list, and other documents listed in the applicable specification and assembly drawings. Any conflict between this specification and drawings, the drawing shall take precedence.

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3.3 General specification. The Control Assembly shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.3.1 MIL-F-13926.

a. Dimensions and tolerances
b. Inorganic protective surface finishes
c. Workmanship
d. Soldering electrical connections

3.3.2 MIL-C-46357.

a. Wiring
b. Threaded fasteners
c. Marking
d. Finishes

3.4 Environmental.

3.4.1 Vibration (nonoperating). The Control Assembly shall be caplable of withstanding vibration test requirements as defined in MIL-STD-810, Method 514, Procedure X, Curve AX of Figure 514.2.7 in each of the three mutually perpendicular axes without being damaged or adversely affected. The duration of vibration in each axis shall be 84 minutes (Table 514.2 - VII). Subsequent to the vibration test, the Control Assembly shall not have been damaged or adversely affected and shall meet all requirements of 3.6.

3.4.2 Humidity (nonoperating). The Control Assembly shall be capable of withstanding the humidity test requirements defined in MIL-STD-810, Method 507.1, without being damaged or adversely affected. After the humidity tests, excess moisture shall be removed from external surfaces and connectors. Subsequent to the humidty test and removal of excess moisture, the Control Assembly shall not have been damaged or adversely affected and shall meet all requirements of 3.6.

3.4.3 Sand and Dust (nonoperating). The Control Assembly shall be capable of withstanding the sand and dust test requirements as defined in MIL-STD-810, Method 510, without being damaged or adversely affected. Subsequent to the sand and dust test, the Control Assembly shall not have been damaged or adversely affected and shall meet all requirements of 3.6 with case closed.

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3.4.4 Altitude (nonoperating). The Control Assembly shall be capable of withstanding the altitude test requirements as defined in MIL-STD-810, Method 500.1, without being damaged or adversely affected. The duration of the test shall be three hours at an altitude of 40,000 feet.Subsequent to the altitude test, the Control Assembly shall not have been damaged or adversely affected and shall meet all requirements of 3.6.

3.4.5 Fungus (nonoperating). The Control Assembly shall be capable of withstanding the fungus test requirements as defined in MIL-STD-810, Method 508, without being damaged or adversely affected. Subsequent to the fungus test, the Control Assembly shall not have been damaged or adversely affected and shall meet all requirements of 3.6.

3.4.6 Temperature (nonoperating). The Control Assembly shall be capable of withstanding the temperature test requirements as defined in MIL-STD-810, Method 501.1 and 501.2 without being damaged or adversely affected. The Control Assembly shall be temperature stabilized at plus 70 $\pm$ 2°C and minus 50 $\pm$ 2°C for a period of 48 hours at each temperature extreme. After each of the above temperature tests, the Control Assembly shall be temperature stabilized at 25 $\pm$ 5°C for at least two hours. Subsequent to the temperature stabilization period, the Control Assembly shall not have been damaged or adversely affected and shall meet all requirements of 3.6 at each temp extreme. Perform electrical tests per Test Plan Figure 4.

3.4.7 Temperature (operating).

3.4.7.1 High temperature. The Control Assembly shall be capable of withstanding the temperature test requirements as defined in MIL-STD-810, Method 501.1, without being damaged or adversely affected. The Control Assembly, after a minimum of three hours temperature stabilization at +57 $\pm$ 2°C with nominal primary power, shall be capable of continuous operation for four hours. During the four hour operating interval, the Control Assembly shall be tested in the first and third hours with nominal primary power. Perform electrical tests per Test Plan Figure 4.

3.4.7.2 Low temperature. The Control Assembly shall be capable of withstanding the temperature test requirements as defined in MIL-STD-810, Method 502.1, without being damaged or adversely affected. The Control Assembly, after a minimum of three hours temperature stabilization at minus 42 $\pm$ 2°C,

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without primary power applied, shall be capable of continuous operation for four hours. During the four hour operating interval, the Control Assembly shall be tested 40 minutes after nominal primary power (+24 ± 1 VDC) is applied and during the third hour of the operating test interval to meet all requirements of 3.6.21.

3.4.8 Salt Fog (nonoperating). The Control Assembly shall be capable of withstanding the salt-fog test requirements as defined in MIL-STD-810, Method 509, without being damaged or adversely affected. Subsequent to the salt-fog test, the Control Assembly shall not have been damaged or adversely affected and shall meet all requirements of 3.6 with case closed.

3.4.9 Rain (nonoperating). The Control Assembly shall be capable of withstanding the rain test requirements as defined in MIL-STD-810, Method 506, Procedure 1, without being damaged or adversely affected. After the rain test, excess moisture shall be removed from external surfaces and connectors. Subsequent to removal of excess moisture, the Control Assembly shall not be damaged or adversely affected and shall meet all requirements of 3.6 with case closed.

3.4.10 Immersion (nonoperating). The Control Assembly case shall be capable of withstanding the immersion test requirements as defined in MIL-STD-810, Method 512.1, without being damaged or adversely affected. The Control Assembly case with equivalent dummy load and pressure release valve replaced with a sealed plug shall be immersed in water and subjected to a positive pressure equivalent to three head-feet of water for a two hour period. Subsequent to the immersion test, the Control Assembly case shall have no evidence of water penetration within the case.

3.5 Electrical.

3.5.1 Continuity. The Control Assembly shall demonstrate continuity in accordance with Drawing 11748460 Wire List.

3.6 Performance. The performance requirements specified herein shall be met at Standard Ambient Conditions on each and every Control Assembly prior to Government acceptance.

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3.6.1 Standard ambient conditions.

a. Temperature +18.2°C to +29.4°C
b. Relative humidity 90 percent RH max
c. Pressure 28 to 32 inches of Hg

3.6.2 Primary Input Power

3.6.2.1 The Control Assembly shall be capable of meeting all requirements of this specification with +24 ± 6 VDC primary input power (maximum current 12 amps).

3.6.3 Power Requirements

3.6.3.1 The Control Assembly shall be operated from a Power Supply Assembly (Unit 2) 11748133 meeting the requirements of MIL-P-48606.

3.6.3.2 The Power Supply Assembly (Unit 2) 11748133 shall not be considered for environmental testing or failure definition purposes relative to this specification.

3.6.3.3 In lieu of the Power Supply Assembly (Unit 2) 11748133, the Control Assembly may be operated from individual power supplies meeting all requirements of MIL-P-48606.

3.6.4 Warm-Up

3.6.4.1 An initial warm up period of 10 to 15 minutes shall occur prior to the commencement of any performance testing unless otherwise specified.

3.6.5 Definitions

3.6.5.1 The term "(CW)" shall define the condition of the Control Assembly in which R.F. switches "S1" in assembly 1A1 (11748168) and the "Gate Switch" in assembly 1A5 (11748455) are in a steady state low insertion condition.

3.6.5.2 The term "(Pulse)" shall define the condition of the Control Assembly in which R.F. switches "S1" in assembly 1A1 (11748168) and the "Gate Switch" in assembly 1A5 (11748455) are being controlled by their respective pulse drivers.

3.6.5.3 The Control Assembly shall be in the pulse condition when cable W2 is connected to Signal Input J3 and shall be in the CW condition when cable W2 is disconnected from Signal Input J3.

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3.6.6 Voltage Monitor

3.6.6.1 The following requirements shall be met with POWER SUPPLY Cable W1 connected to Control Assembly connector J2 and POWER INPUT Cable W3 connected to Control Assembly connector J4.

3.6.6.2 With the FUNCTION SELECT switch in the VOLTAGE position and the VOLTAGE SELECT switch in the positions delineated in Table I, the Digital Display (DVM) shall indicate the allied voltage delineated in Table I within a tolerance of ± 0.050 VDC.

TABLE I

VOLTAGE SELECT (Position)	APPLIED VOLTAGE Value (VDC)	Applied
+24	+24 ± 0.200	W3-A (+) W3-B (ret.)
+ 5	+ 5 ± 0.200	W1A (+) W1-B (ret.)
+15	+15 ± 0.300	W1-E (+) W1-F (ret.)
-15	-15 ± 0.300	W1-G (-) W1-F (ret.)
+28	+28 ± 0.600	W1-C (+) W1-D (ret.)

3.6.6.3 The voltage at cable end W1 between W1-H (ret.) and W1-I (+) shall be as defined in para. 3.6.6.2 for the +24 VOLTAGE SELECT (position).

3.6.7 Voltage Input Monitor.

3.6.7.1 With the FUNTION SELECT switch in VOLT the VOLTAGE position and the VOLTAGE SELECT switch in the VOLT TP position, the Digital Display (DVM) shall indicate the applied inputs delineated in Table II when applied to the VOLT TP input jacks. (Applied Input - INPUT, Ret - GND) Note - Do not exceed 36 volts.

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TABLE II

APPLIED INPUT	DIGITAL DISPLAY (DVM)
+35.00 ± 00.10	+35.00 ± 00.15
Short	0.000 ± 0.010
Open	0.000 ± 0.015
-35.00 ± 00.10	-35.00 ± 00.15
+00.10 ± 00.01	+00.10 ± 00.02

3.6.8 Stalo Output.

3.6.8.1 All requirements shall be measured at the STALO Out port with the MODE SELECT switch in the RADAR position unless otherwise indicated.

3.6.8.1.1 The Control Assembly shall be in the (Pulse) condition.

3.6.8.2 The R.F. output power shall be +12 dbm minimum for all positions of the FREQ SELECT switch.

3.6.8.3 The R.F. output power shall be attenuated a minimum of 30 db relative to the R.F. output power (para. 3.6.8.2) with the MODE SELECT switch in the SELF TEST position for all positions of the FREQ SELECT switch.

3.6.8.4 The Frequency stability shall be as defined in Table III.

TABLE III

FREQ SELECT Switch (Front Panel)	Frequency Stability (± 0.5 MHz)
9.205	9,145 MHz
9.213	9,153 MHz
9.221	9,161 MHz
9.229	9,169 MHz
9.237	9,177 MHz
9.245	9,185 MHz

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3.6.8.5 The following performance requirements of MIL-S-48431 as defined in Table IV shall be met for all positions of the FREQ SELECT switch.

TABLE IV

Specification	Paragraph	Title
MIL-S-48431	3.4.6	Spectral Purity
MIL-S-48431	3.4.6.1	Power Source Noise Spectral Density (Broadband)

3.6.9 Self Test Signal

3.6.9.1 All requirements shall be measured at the CAL OUT port with the MODE SELECT switch in the SELF TEST position unless otherwise indicated.

3.6.9.1.1 The Control Assembly shall be in the (CW) condition unless otherwise indicated.

3.6.9.2 The R.F. output power shall be -7 dbm minimum for all positions of the FREQ SELECT switch.

3.6.9.3 The R.F. output power shall be attenuated a minimum of 80 db relative to the R.F. output power (Para. 3.6.9.2) with the MODE SELECT switch in the RADAR position for all positions of the FREQ SELECT switch.

3.6.9.4 The frequency stability shall be as defined in Table V.

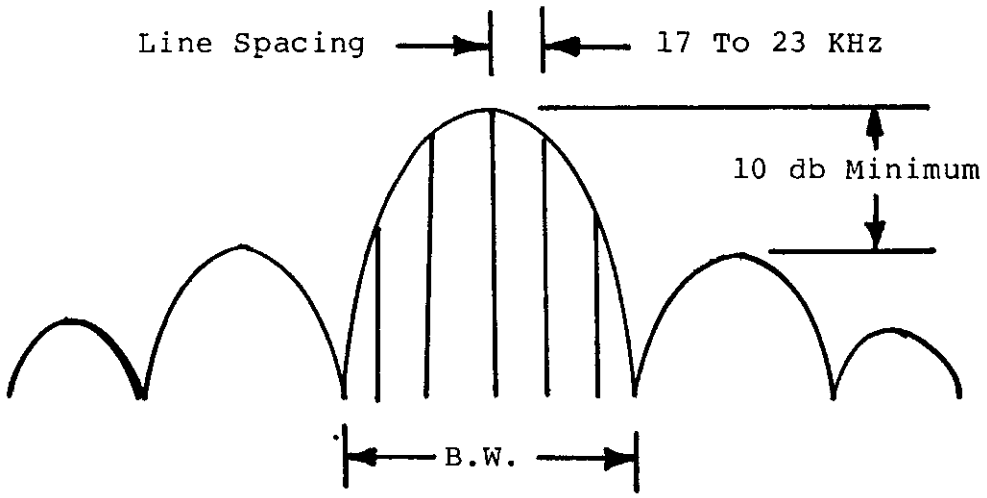
TABLE V

FREQ SELECT Switch (Front Panel)	Frequency Stability (± 1.0 MHz)
9.205	9,205 MHz
9.213	9,213 MHz
9.221	9,221 MHz
9.229	9,229 MHz
9.237	9,237 MHz
9.245	9,245 MHz

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3.6.9.5 All harmonic and spurious R.F. power shall be attenuated a minimum of 20 db relative to the R.F. output power (Para. 3.6.9.2) for all positions of the FREQ SELECT switch. Note: measurement bandwidth shall be the frequencies (Para. 3.6.9.4) ± 180 MHz

3.6.9.6 With the Control Assembly in the (Pulse) condition, the pulse spectrum for all positions of the FREQ SELECT switch shall be as defined in Figure 1.



PULSE SPECTRUM

FIGURE I

3.6.9.7 With the CONTROL ASSEMBLY IN THE (Pulse) condition and the FUNCTION SELECT switch in the R.F. INPUT position, the R.F. LEVEL attenuator shall be capable of adjusting the Self Test Signal such that the Digital Display (DVM) reads ± 1.000 ± 0.020 for all positions of the FREQ SELECT switch.

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3.6.9.8 With the FUNCTION SELECT switch in the R.F. INPUT position and the R.F. LEVEL attenuator dial set to 0 (zero), the Digital Display (DVM) shall indicate (+ or -) 0.020 ± 0.030 for all positions of the FREQ SELECT switch.

3.6.9.8.1 With MODE SELECT switch in the RADAR position, the requirements delineated above (Para. 3.6.9.8) are also applicable.

3.6.10 Cal Input

3.6.10.1 The attenuation between the CAL IN port and the CAL OUT port shall be less than 6 db over the frequency range from 9,205 to 9,245 MHz.

3.6.11 R.F. Monitor

3.6.11.1 All requirements shall be met for all positional combinations of the FREQ SELECT switch and the RANGE SELECT switch unless otherwise indicated.

3.6.11.2 The FUNCTION SELECT switch shall be in the R.F. MONITOR position.

3.6.11.3 The Control Assembly shall be in the (Pulse) condition unless otherwise indicated.

3.6.11.4 With the MODE SELECT switch in the SELF TEST position, the R.F. LEVEL attenuator shall be capable of adjustment such that the Digital Display (DVM) reads +2.500 ± 0.050.

3.6.11.4.1 The R.F. LEVEL attenuator dial shall indicate 8 db or greater at ambient conditions and 3 db or greater at any other environmental condition specifically required by this specification.

3.6.11.5 With the MODE SELECT switch in the RADAR position and the R.F. LEVEL attenuator dial at 0 (zero), the Digital Display (DVM) shall read (+ or -) 0.020 ± 0.030.

3.6.11.6 With the MODE SELECT switch in the SELF TEST position, the R.F. LEVEL attenuator shall be adjusted such that the Digital Display (DVM) indicates +2.500 ± 0.020. After the initial R.F. LEVEL attenuator adjustment has been accomplished, the Digital Display (DVM) shall remain stable within a reading of +2.500 ± 0.070 for a duration of 90 ± 10 seconds.

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3.6.11.6.1 The above requirements (Para. 3.6.11.6) shall utilize the FREQ SELECT and RANGE SELECT combinations given in Table VI.

TABLE VI

FREQ SELECT Position	RANGE SELECT Position
9.205	R1
9.213	R2
9.221	R3
9.229	R4
9.237	R4
9.245	R3

3.6.11.7 With the Control Assembly in the (CW) condition and the R.F. LEVEL attenuator set to O (Zero), the Digital Display (DVM) shall indicate (+ or -) 0.020 ± 0.030 for all positions of the FREQ SELECT switch and RANGE SELECT positions as given in Table VI.

3.6.12 Video Output

3.6.12.1 The Control Assembly shall be in the (Pulse) condition.

3.6.12.2 All requirements shall be measured at the VIDEO OUT port with the MODE SELECT switch in the SELF TEST position, the function select switch in the R.F. MONITOR position, and the R.F. LEVEL attenuator adjusted such that the Digital Display (DVM) indicates 2.500 ± 0.100 unless otherwise indicated.

3.6.12.3 The video output shall be a pulse having a minimum amplitude of 1.0V (peak-to-peak) and a pulse repetition frequency between 17 KHz and 23 KHz for all positions of the FREQ SELECT switch.

3.6.12.4 With the MODE SELECT switch in the RADAR position and the R.F. LEVEL attenuator dial set to 0 (zero), the video output shall be less than 100 MV (peak-to-peak) for all positions of the FREQ SELECT switch.

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3.6.13 Data Signals

3.6.13.1 The following requirements shall be met with DATA INPUT cable W4 connected to Control Assembly connector J1. The applied voltages shall be applied to the other end of cable W4.

3.6.13.2 With the FUNCTION SELECT switch in the positions delineated and applied voltages delineated in Table VII, the Digital Display (DVM) shall display the values delineated in Table VII.

TABLE VII

FUNCTION SELECT (Position)	APPLIED VOLTAGE Value (VDC)	Applied	DIGITAL DISPLAY (DVM)
RANGE RATE REF (+)	±10.00 ± 00.10	K (+) F (RET)	+10.00 ± 00.15
RANGE REF	+20.00 ± 00.10	J (+) B (RET)	+20.00 ± 00.15
RANGE	+20.00 ± 00.10	A (+) B (RET)	+20.00 ± 00.15
RANGE RATE	+10.00 ± 00.10	H (+) F (RET)	+10.00 ± 00.15
RANGE RATE REF (-)	-10.00 ± 00.10	G (-) F (RET)	-10.00 ± 00.15

3.6.14 Signal Channel

3.6.14.1 The following requirements shall be met with SIGNAL INPUT cable W2 connected to Control Assembly connector J3.

3.6.14.2 With a 10 ± 0.1 KHz sine wave signal with a PK to PK amplitude of 1.0 ± 0.1 volt applied to open end of SIGNAL INPUT cable W2; Signal-B, Ret-A; the SIGNAL CHANNEL meter shall indicate 0.5 volts min.

3.6.15 Lock On Signal

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3.6.15.1 The following requirements shall be met with DATA INPUT cable W4 connected to Control Assembly connector J1.

3.6.15.2 With Applied signals to open end of DATA INPUT cable W4; Signal-C, Ret-0; as delineated in Table VIII, the LOCK ON indicator shall indicate as delineated in Table VIII.

TABLE VIII

APPLIED SIGNAL	LOCK ON INDICATOR
+5.0 ± 0.1 VDC	Not Lighted
+5.0 ± 0.1 VDC	Lighted (Indicator Depressed)
Open	Lighted
+1.0 ± 0.1 VDC	Lighted

3.6.16 Elapse Time Indicator

3.6.16.1 With POWER SUPPLY cable W1 connected to Control Assembly connector J2 and +28 ± 0.6 VDC applied to cable end W1; W1-C (+), W1-D (Ret); the elapsed time indicator shall have an audio sound indicating activation and shall display operational time within 33 percent in a three hour continuous period.

3.6.17 Range and Phase Insertion

3.6.17.1 The Control Assembly shall be in the (Pulse) condition.

3.6.17.2 With the FUNCTION SELECT switch in the R.F. MONITOR position, the RANGE SELECT switch in the R3 position, and the PHASE CONTROL at midrange; the R.F. LEVEL attenuator shall be adjusted such that the Digital Display (DVM) indicates +2.500 ± 0.020.

3.6.17.3 Subsequent to the requirements delineated in paragraph 3.6.17.1, the Digital Display (DVM) shall not vary more than ± 1.200 for RANGE SELECT switch positions R1, R2, and R4.

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3.6.17.4 Subsequent to the requirements delineated in paragraph 3.6.17.1, the Digital Display (DVM) shall not vary more than + 0.500 when the PHASE CONTROL is varied from one extreme (CCW) position to the other extreme (CW) position.

3.6.17.5 All requirements shall be measured for all positions of the FREQ SELECT switch.

3.6.18 Clutter Signal Characteristics

3.6.18.1 All clutter signal characteristics are to be measured with an internal R.F. power calibration (Pulse) accomplished prior to (within 30 seconds) of any measurement unless otherwise indicated.

3.6.18.2 An internal R.F. power calibration (Pulse) shall be accomplished by adjusting the R.F. LEVEL dial such that the Digital Display (DVM) indicates +2.500 + 0.025 with the R.F. FUNCTION switch in the R.F. MONITOR position.

3.6.18.3 All clutter signal characteristics shall be measured with the MODE SELECT SWITCH IN THE SELF TEST position unless otherwise indicated.

3.6.18.4 Unless otherwise indicated, all clutter signal characteristics shall be measured for all positions of the FREQ SELECT switch with the Control Assembly in the (CW) condition.

3.6.18.5 All clutter signal characteristics shall be measured at one end of cable W5 with the other end of cable W5 connected to the RADAR port of the Control Assembly unless otherwise indicated.

3.6.18.6 With the CLUTTER switch in the OFF position, CLUTTER LEVEL dial set to 5 (five), and the CLUTTER BALANCE attenuator set to minimum; the clutter signal (CW) shall be less than -90 dbm.

3.6.18.7 With the CLUTTER switch in the ON position and the CLUTTER LEVEL dial set to 20, the CLUTTER BALANCE attenuator shall be capable of adjusting the clutter signal (CW) from -18 dbm to -22 dbm.

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3.6.18.8 All harmonically related clutter power shall be attenuated a minimum of 40 db relative to the clutter signal (CW) over a frequency range of (clutter signal ± 180 MHz).

3.6.18.9 All spurious related clutter power shall be attenuated a minimum of 60 db relative to the clutter signal (CW) over a frequency range of (clutter signal ± 180 MHz).

3.6.18.10 The clutter signal (CW) shall be attenuated referenced to the CLUTTER LEVEL dial over the dial range from 5 to 65 within the accuracy requirements given in table IX . An internal R.F. power calibration (pulse) is not required. FREQ SELECT switch positions 9.205 and 9.245 only.

TABLE IX

CLUTTER		TARGET	
DIAL READING IN DB	ROOM TEMP. TOLERANCE IN DB	DIAL READING IN DB	ROOM TEMP. TOLERANCE IN DB
15	± 1.5	37	± 1.5
25	± 1.5	47	± 1.5
35	± 1.5	57	± 1.5
45	± 1.5	67	± 1.5
55	± 1.5	77	± 1.5
65	± 2.0	87	± 1.5

TABLE X

ATTENUATION PARAMETER	ACCURACY
Absolute	± 0.3 db ± 0.005 (A-10) db A=10 Temperature 23 ± 2°C
Temperature	± 0.015 db/°C
Frequency	± 0.1 db Frequency 9,205 to 9,245 MHz

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3.6.18.11 The clutter signal (CW) shall be 60 ± 1.0 MHz higher in frequency than the stalo output frequency defined in para. 3.6.8.4

3.6.18.12 With the CLUTTER LEVEL dial at 20, cable W5 connected to the SELF TEST port of the Control Assembly, and CLUTTER BALANCE attenuator adjusted such that with the FUNCTION SELECT switch in the R.F. SELF TEST position the Digital Display (DVM) indicates the R.F. Self Test Calibration value (Pulse); the clutter signal (CW) shall be -20.0 ± 0.1 DBM. FREQ SELECT switch positions 9.229, 9.205, and 9.245 only.

3.6.18.13 With the CLUTTER LEVEL dial set at 5 (five), an increase of 5 (higher dial setting) in the R.F. LEVEL dial from its internal R.F. power calibration setting shall cause a 5 ± 1.0 db reduction in the clutter signal (CW).

3.6.18.14 The clutter signal (Pulse) shall have a pulse spectrum as delineated in Figure 1 with the R.F. LEVEL dial set at its internal R.F. power calibration (Pulse) reference setting and shall maintain the pulse spectrum characteristics when the R.F. LEVEL dial is increased (higher dial setting) an increment of 5.

3.6.19 Target Signal Characteristics

3.6.19.1 All target signal characteristics are to be measured with an internal R.F. power calibration (Pulse) accomplished prior to (within 30 seconds) of any measurement unless otherwise indicated.

3.6.19.2 An internal R.F. power calibration (Pulse) shall be accomplished by adjusting the R.F. LEVEL attenuator such that the DIGITAL DISPLAY (DVM) indicated +2.500 ± 0.025 with the R.F. FUNCTION switch in the R.F. MONITOR position.

3.6.19.3 All target signal characteristics shall be measured with the MODE SELECT switch in the SELF TEST position unless otherwise indicated.

3.6.19.4 Unless otherwise indicated, all target signal characteristics shall be measured for all positions of the FREQ SELECT switch with the Control Assembly in the (CW) condition.

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3.6.19.5 All target signal characteristics shall be measured at one end of cable W5 with the other end of cable W5 connected to the RADAR port of the Control Assembly unless otherwise indicated.

3.6.19.6 With the TARGET switch in the OFF position, TARGET LEVEL attenuator dial set to 27, and the TARGET BALANCE attenuator set to minimum, the target signal (CW) shall be less than -90 dbm.

3.6.19.7 With the TARGET switch in the ON position and the TARGET LEVEL attenuator dial set to 27, the TARGET BALANCE attenuator shall be capable of adjusting the target signal (CW) from -22 dbm to -28 dbm.

3.6.19.8 The target (main spectral line) signal (CW) shall be 3.5 KHz to 5.5 KHz higher in frequency than the clutter signal (para. 3.6.18.11).

3.6.19.9 The target spectrum (CW) within a 200 KHz bandwidth shall have all spurious and harmonically related sidebands suppressed a minimum of 20 db from the target signal (main spectral line).

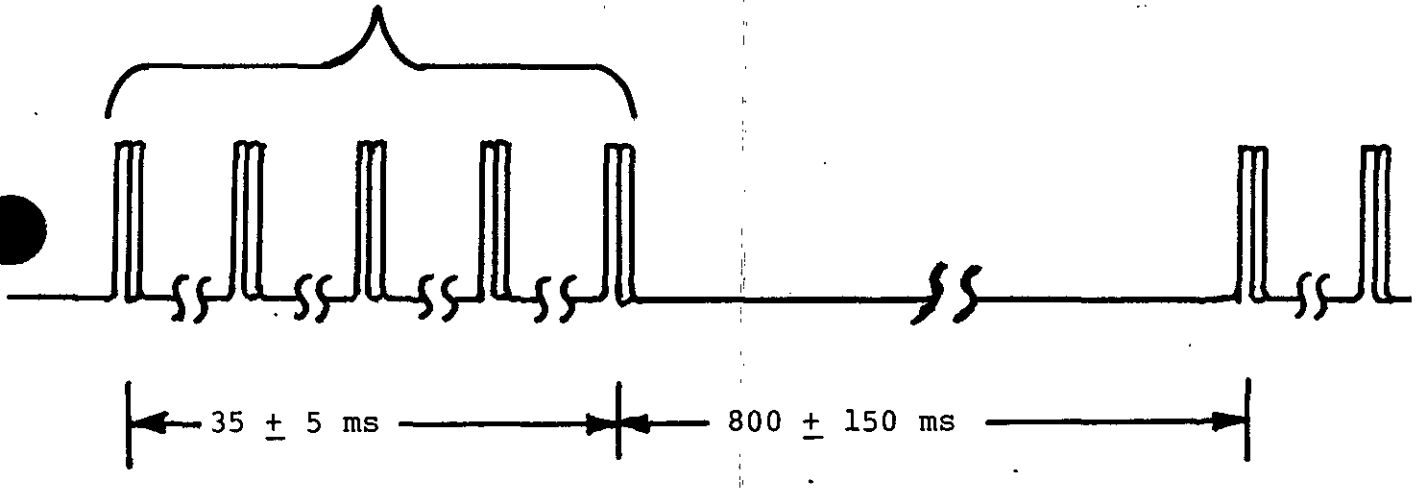
3.6.19.10 The target signal (CW) shall be attenuated referenced to the TARGET LEVEL attenuator dial over the dial range from 27 to 87 within the accuracy requirements given in Table X. An internal R.F. power calibration (pulse) is not required. FREQ SELECT switch in 9.205 and 9.245 positions.

3.6.19.11 With the TARGET switch and the TARGET FREEZE switch in the positions indicated, the target signal when envelope detected shall be as indicated in Table XI. FREQ SELECT switch in the 9.229 position.

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TABLE XI

TARGET Switch	TARGET FREEZE Switch	TARGET SIGNAL OUTPUT
OFF	OFF	None (Less than 10%)
OFF	ON	None (Less than 10%)
ON	OFF	Pulse (100% amplitude ref.)
ON	ON	Figure 2



TARGET SIGNAL OUTPUT

FIGURE 2

3.6.19.12 With the CLUTTER LEVEL attenuator dial at 25 (subsequent to the accomplishment of the requirement delineated in para. 3.6.18.12), the TARGET LEVEL attenuator dial at 27, cable W5 connected to the SELF TEST port of the control assembly, and the FUNCTION SELECT switch in the R.F. SELF TEST position; the TARGET BALANCE attenuator shall be capable of adjustment such that the Digital Display (DVM) indicates the same value within ± 0.010 for the target signal (Pulse) as the clutter signal (Pulse).

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3.6.19.12.1 Subsequent to the accomplishment of the requirements delineated in para. 3.6.19.12, the difference between the target signal (CW) and the clutter signal (CW) shall be 1.5 db max.

3.6.19.12.2 Subsequent to the accomplishment of the requirements delineated in para. 3.6.19.12, the clutter signal shall vary less than ± 1.0 db with the TARGET switch ON or OFF.

3.6.19.13 With the TARGET LEVEL attenuator dial set at 27, an increase of 5 (higher dial setting) in the R.F. LEVEL attenuator dial setting from its internal R.F. power calibration setting shall cause a 5 ± 1.0 db reduction in the target signal (CW).

3.6.19.14 The target signal (Pulse) shall have a pulse spectrum as delineated in Figure 1 with the R.F. LEVEL dial set at its internal R.F. power calibration (Pulse) reference setting and shall maintain the pulse spectrum characteristics when the R.F. LEVEL dial is increased an increment of 5 (higher dial setting).

3.6.20 Calibration Data

3.6.20.1 R.F. INPUT calibration data shall be determined such that when an R.F. input peak power between 0.125 KW and 2.0 KW is applied to the end of cable W5 with the other end of cable W5 connected to the Control Assembly RADAR port, the FUNCTION SELECT switch in the R.F. INPUT position, and the R.F. LEVEL attenuator adjusted such that the Digital Display (DVM) indicates +2.500 ± 0.020; that the R.F. INPUT calibration number when added to the R.F. LEVEL attenuator dial setting measures the R.F. input peak power within ± 0.5 db for R.F. input frequencies between 9,205 and 9,245 MHz with a PRF of 20.0 ± 0.2 KHz and a pulse width of 280 ± 20 ns.

3.6.20.1.1 The R.F. INPUT calibration value shall be within the limits of 30.0 and 52.0.

3.6.20.1.2 The R.F. INPUT cablibration data is to be a single value in db (xx.x) applicable for all conditions and shall be recorded on the calibration data card located in the Control Assembly panel flyleaf.

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3.6.20.2 R.F. SELF TEST calibration data shall be determined such that with a -20.0 ± 0.1 dbm peak power injected directly into the Control Assembly SELF TEST port, the Digital Display (DVM) shall indicate the R.F. SELF TEST calibration value within ± 0.100 with the FUNCTION SELECT switch in the R.F. SELF TEST position. The injected R.F. peak power shall have the following characteristics: frequency 9,229 MHz, PRF 17 to 23 KHz, and pulse width 300 ± 50 ns.

3.6.20.2.1 The R.F. SELF TEST calibration value shall be within the limits of 1.50 and 2.50.

3.6.20.2.2 The R.F. SELF TEST calibration data is to be a single value Digital Display (DVM) indication (x.xx) applicable for all conditions and shall be recorded on the calibration data card located in the Control Assembly panel flyleaf.

3.6.20.3 RANGE calibration data shall be determined for RANGE SELECT switch positions R1, R2, R3 and R4. The range delay (pulse delay) shall be determined in nanoseconds for an R.F. pulse starting at the CAL OUT port and stopping at the open end of cable W5 (other end of cable W5 connected to RADAR port). The range delay shall be measured with the CLUTTER switch in the ON position and the TARGET switch in the OFF position. The R.F. input (start) pulse shall have the following characteristics: frequency 9,205 MHz to 9,245 MHz, PRF 17 KHz, to 23 KHz, pulse width 250 ns to 350 ns.

3.6.20.3.1 The range delay shall vary no more than ±25 nanoseconds from the RANGE calibration data in nanoseconds for all conditions.

3.6.20.3.2 The range delay for R1, R2, R3, and R4 shall be converted from nanoseconds to meters using 0.15 meter equals 1.0 nanoseconds.

3.6.20.3.3 The RANGE calibration data shall be within the following limits: R1 300 ± 20, R2 400 ± 20, R3 1575 ± 20, and R4 4715 ± 20.

3.6.20.3.4 The range delay in meters (xxxx.) shall be recorded as the RANGE calibration data on the calibration data card located in the Control Assembly panel flyleaf.

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3.6.21 Radar Test

3.6.21.1 The Control Assembly shall be capable of performing a complete RADAR TEST by measuring the following parameters of an AN/VPS-2 Vulcan Radar as given in table XII for all positions of the FREQ SELECT switch unless otherwise indicated.

TABLE XII

AN/VPS-2 PARAMETER	ROOM TEMP. ACCURACY	ACCURACY OVER TEMP. RANGE
R.F. Peak Input Power	± 1.0 dbm	± 1.5 dbm
Minimum Lock On Signal (MDS)	± 1.0 dbm	± 1.5 dbm
Sub Clutter Visability (SCV)	± 1.5 db	± 2.0 db
AGC Cut In	± 1.0 db	± 1.5 db
AGC Linearity	± 1.5 db	± 2.0 db
Range (R1, R2, R3, R4)	± 7 Meters	±12 Meters
Range Rate Ref (+/-)	± 0.050 VDC	± 0.075 VDC

3.6.21.1.1 During enviromental operating test (para. 3.4.5 and 3.4.9) and reliability run-in enviromental operating tests (para. 3.7.1) the FREQ SELECT switch shall be in the 9.205 and 9.245 positions.

3.6.21.2 If for ANY reason an AN/VPS-2 Vulcan Radar is not available then in lieu of the requirements delineated in para. 3.6.21.1 all other requirements of para. 3.6 PERFORMANCE shall be accomplished.

3.6.21.2.1 If for any reason an An/VPS-2 Vulcan Radar is not available, then during environmental operating tests (Para. 3.4.5 and 3.4.9) and reliability run-in environmental operating tests (para. 3.7.1), in lieu of the requirements delineated in para. 3.6.21.1, all other requirements of para. 3.6 PERFORMANCE shall be accomplished with the exception of para. 3.6.8.5

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3.6.21.3 The AN/VPS-2 Vulcan Radar shall be maintained in a calibration status by external independent verification of parameter data, by periodic verification utilizing the factory test rack, or by verification of calibration status utilizing an AN/VPM-2 system having met all requirements of MIL-P-48606 and MIL-C-48605.

3.7 Reliability Run-In - In order to maximize component reliability and prior to acceptance by the Government, each and every Control Assembly shall be successfully subjected to the following Reliability Run-In requirements of Figure 5.

3.7.1 Failures during Reliability Run In - Replacement of defective Power Modules and correction of workmanship defects may be made during reliability run-in under the following conditions:

a. Workmanship corrections can be made with no additional testing required.

b. Module Replacement - Replacement Power Modules must be subjected to the requirements of Figure 5 up to the point of failure. After assembly of the replacement module is complete, the testing is to be resumed at the beginning of the test where the failure occured and continued until the requirements of Figure 5 are complete.

4. QUALITY ASSURANCE PROVISIONS

4.1 General

4.1.1 Responsibility for Inspection. - Unless otherwise specified in the contract, the supplier is responsible for the performance of all test and inspection requirements specified herein. Except as otherwise specified in the contract the supplier will specify facilities suitable for the performance of the test and inspection, unless disapproved by the Government. The Government reserves the right to perform any of the tests and inspections deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1.1 General Provisions - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

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4.1.2 Special Tests and Examinations.

4.1.2.1 First Article (initial production) Approval - The requirements for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall be 3 assemblies manufactured in the same manner using the same mataerials, equipment, processes, including packaging and packing, obtained from the same source of supply as used in regular production. The assemblies shall meet the requirements and tests of Figure 4.

4.1.2.1.1 Government Testing - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.1.2.1.2 Contractor Testing - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.1.2.2 Control Tests - Three assemblies shall be selected at random as a control sample from each 50 produced. The control samples shall meet the following requirements and tests of Figure 4.

a. Design and construction
b. Vibration
c. Temperature, operating
d. Performance

4.1.2.3 Failure of Sample - Should any one item of a control sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.1.2.2).

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4.2 Quality Conformance Inspections

4.2.1 Submission of Product - Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Acceptance and Rejection - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and re-submittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.3 Inspection Equipment - Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.2.3.1 Contractor Test Equipment - The contractor shall provide test and inspection equipment compatible with the "Test Methods and Procedures" specified in the detail specification and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerance used in the test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerances limit, and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.2.3.1.1 Test Equipment

1. Power Supply (Unit 2) 11748133

2. Power Supply, 0 to 30 Volts, 15 Amps min. HP Model #6456 or equivalent

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3. Cables:

W1 - 11748213
W2 - 11748214
W3 - 11748215
W4 - 11748216
W5 - 11748144

4. Power Supply, 0 to 40 Volts, 0.5 Amps

5. Function Generator - Systron Donner, Model #415 or equivalent

6. Frequency Counter, HP 5345A with X-Band Plug-In or equivalent

7. Two RF Amplifiers, Part No. 11748155

8. X-Band Rotary Vane Waveguide Variable Attenuator, Waveline Model #60562 or equivalent

9. Two coaxial Detectors, Aertech Model #8596BZS or equivalent

10. Waveguide to Coaxial Adapter, Omin Spectra Model #2000-6254-00 or equivalent

11. Precision Attenuator Set - Weinschel Engineering, Model #9168 or equivalent

12. Spectrum Analyzer, Tectronics, Model #491 or equivalent

13. Power Meter, HP 432A or equivalent

14. Oscilloscope, Tektronics, Model #454 or equivalent

15. Signal Generator, X-Band, Alfred Model #650 with Model #655K plug-in or equivalent

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4.2.4 Examinations - Unless otherwise specified, components and subassemblies shall be inspected in accordance with the "Inspection provisions for components and subassemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations to verify requirements of 3.2 shall be performed. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.5 Functional Tests - Functional requirements shall be inspected on a 100 percent basis. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	18.2°C to 29.4°C
b.	Relative humidity	90 percent RH max.
c.	Pressure	28 to 32 inches of Hg

4.3 Environmental Test Methods and Procedures

4.3.1 Environmental - Test facilities utilized in the following environmental tests shall be in accordance with "test facilities" requirements of MIL-F-13926 and test conditions defined in 3.4.

4.3.1.1 Vibration - The vibration test shall be performed in accordance with MIL-STD-810, Method 514, Procedure X, Curve AX of Figure 514.2.7 and conditions specified in 3.4.1.

4.3.1.2 Humidity - The humidity test shall be performed in accordance with MIL-STD-810, Method 507.1 and conditions specified in 3.4.2.

4.3.1.3 Sand and Dust - The sand and dust test shall be performed in accordance with MIL-STD 810, Method 510, and condition specified in 3.4.3.

4.3.1.4 Altitude

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4.3.1.4.1 Nonoperating - The nonoperating altitude test shall be performed in accordance with MIL-STD-810, Method 500.1 and the conditions specified in 3.4.4.

4.3.1.5 Fungus - The fungus test shall be performed in accordance with MIL-STD-810, Method 508, and conditions specified in 3.4.5.

4.3.1.6 Temperature

4.3.1.6.1 Storage - The storage temperature test shall be performed in accordance with MIL-STD-810, Method 501.1, 501.2, and conditions specified in 3.4.6.

4.3.1.6.2 Operating Temperatures - The operating temperature test shall be performed in accordance with MIL-STD-810, Method 501.1, and conditions specified in 3.4.7.

4.3.1.7 Salt-Fog - The salt-fog test shall be performed in accordance with MIL-STD-810C, Method 509 and conditions specified in 3.4.8.

4.3.1.8 Rain - The rain test shall be performed in accordance with MIL-STD-810C, Method 506, Procedure 1, and conditions specified in 3.4.9.

4.3.1.9 Immersion - The immersion test shall be performed in accordance with 3.3.12. If penetration occurs, the Control Assembly case shall be re-immersed and a negative pressure equivalent to three head-feet of water shall be applied. The Control Assembly case shall be observed for escaping air bubbles to determine penetration points.

4.4 Electrical

4.4.1 Continuity - Continuity shall be tested by standard measuring equipment for compliance with 3.5.1.

4.5 Performance

4.5.1 Initial Conditions

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4.5.1.1 Connect cables W1, W1, W3, W4, and W5 as shown in Figure 3. If individual power supplies are utilized (para. 3.6.3.3) connect in lieu of Power Supply Unit 2 in Figure 3.

4.5.1.2 Set Control Assembly front panel controls as delineated in Table XIII.

TABLE XIII

CONTROL	POSITION
Power	On
Function Select	Voltage
Voltage Select	+24
RF Level	20 db
Freq Select	9.205
Mode Select	Self Test
Phase Control	Mid Range
Clutter	Off
Clutter Level	20 db
Range Select	R1
Target Level	0 db
Target Freeze	Off
Target	Off

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4.5.1.3 Place the Control Assembly into the (CW) condition as defined in para. 3.6.5 unless otherwise indicated.

4.5.1.4 Apply +24 ± 2.0 VDC (unless otherwise specified) to power input cable W3 (either 5J1 or 5P1) pins A (+), B(ret), C(Gnd).

4.5.1.5 Turn POWER switch to ON position.

4.5.1.6 Allow an initial warm-up as defined in para. 3.6.4.

4.5.2 Voltage Monitor

4.5.2.1 Perform initial conditions per para. 4.5.1.

4.5.2.2 Place the FUNCTION SELECT switch in the VOLTAGE position.

4.5.2.3 Measure and record the applied voltage as delineated in Table I.

4.5.2.4 Measure and record the Digital Display (DVM) indication for each position of the VOLTAGE SELECT switch per Table I.

4.5.2.5 The difference between the applied voltage and the Digital Display (DVM) indication is the tolerance and shall be as delineated in para. 3.6.6.2.

4.5.2.6 Measure and record the voltage between W1-H (+) and W1-I (ret) per para. 3.6.6.3

4.5.3 Voltage Input Monitor

4.5.3.1 Perform initial conditions per para. 4.5.1.

4.5.3.2 Place the FUNCTION SELECT switch in the VOLTAGE position.

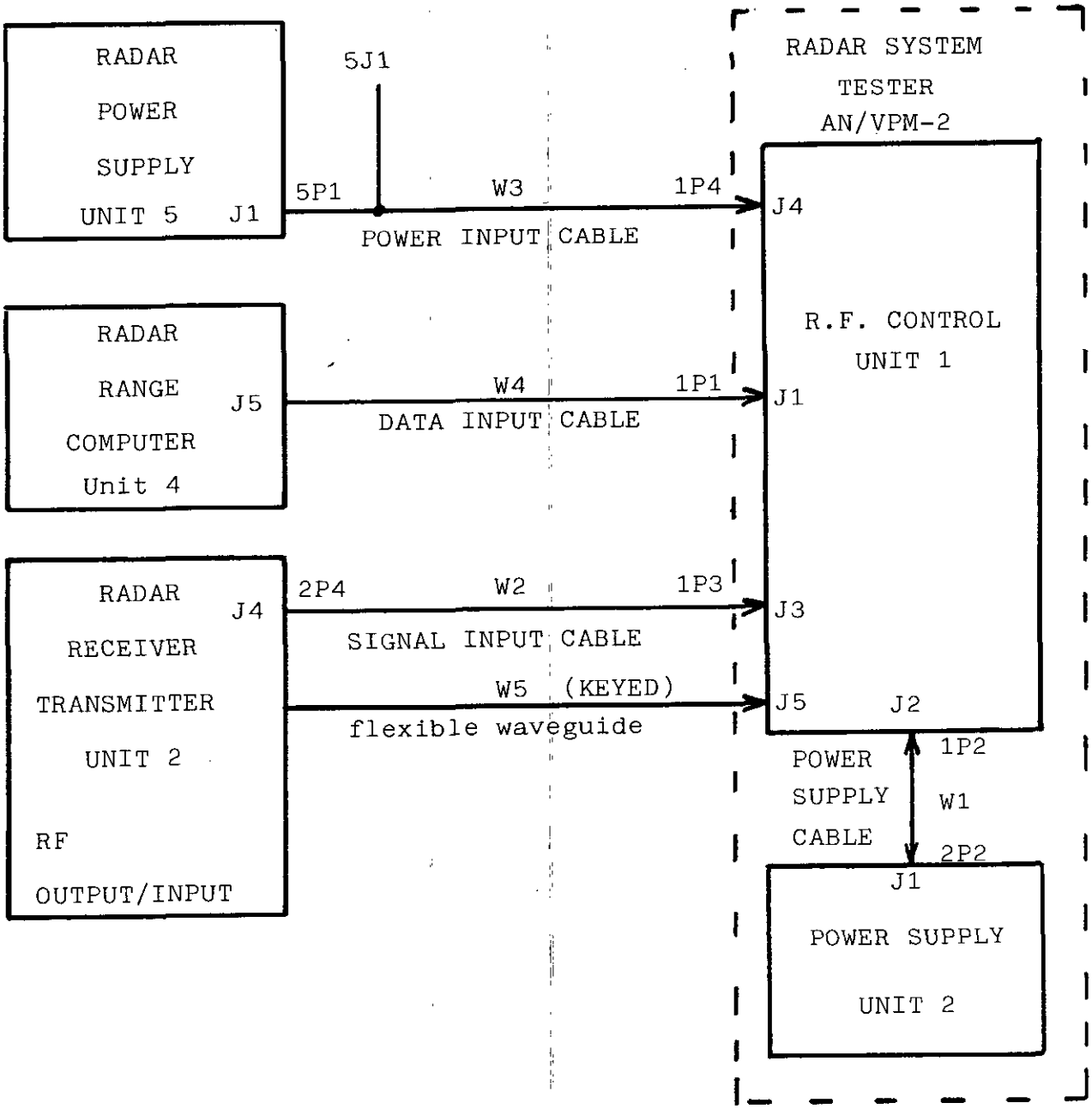
4.5.3.3 Place the VOLTAGE SELECT switch in the VOLT TP position.

4.5.3.4 Apply inputs as delineated in Table II to the INPUT jack and GND jack (para. 3.6.7.1)

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FIGURE 3

RADAR SYSTEM TESTER AN/VPM-2



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4.5.3.5 Measure and record the applied inputs and Digital Display (DVM) indications per Table II.

4.5.4 Stalo Output

4.5.4.1 Perform initial conditions per para. 4.5.1

4.5.4.2 Place the Control Assembly in the (Pulse) condition per para. 3.6.5.

4.5.4.3 Remove the 50 OHM termination from STALO OUT port and connect in its place a 10 db coaxial attenuator and an R.F. power meter, set mode select switch to RADAR position.

4.5.4.4 Measure and record the R.F. output power for all positions of the FREQ SELECT switch per para. 3.6.8.2.

4.5.4.5 Remove the R.F. power meter and connect a 30 db coaxial attenuator and a spectrum analyzer.

4.5.4.6 Measure and record the attenuation with the MODE SELECT switch in the SELF TEST position relative to the RADAR position for all positions of the FREQ SELECT switch per para. 3.6.8.3.

4.5.4.7 Remove the spectrum analyzer and 30 db coaxial attenuator and connect a 10 db coaxial attenuator and frequency counter. Place the MODE SELECT switch in the RADAR position.

4.5.4.8 Measure and record the stalo output frequency for all positions of the FREQ SELECT switch per para. 3.6.8.4 (Table III).

4.5.4.9 Remove the frequency counter and connect test equipment as delineated in MIL-S-48431.

4.5.4.10. Measure and record the spectral purity, power source noise spectral density as required in para. 3.6.8.5 (Table IV).

4.5.4.11 Remove test equipment from STALO OUT port and reconnect 50 OHM termination.

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4.5.5 Self Test Signal

4.5.5.1 Perform initial conditions per para. 4.5.1.

4.5.5.2 Place the Control Assembly in the (CW) condition per para. 3.6.5.3.

4.5.5.3 Remove the 50 OHM termination from the CAL OUT port and connect in its place an R.F. power meter with appropriate adapters.

4.5.5.4 Measure and record the R.F. output power for all positions of the FREQ SELECT switch per para. 3.6.9.2.

4.5.5.5 Remove the R.F. power meter and connect a spectrum analyzer.

4.5.5.6 Measure and record the attenuation with the MODE SELECT switch in the SELF TEST position relataive to the RADAR position for all positions of the FREQ SELECT switch per para. 3.6.9.3.

4.5.5.7 Place the MODE SELECT switch in the SELF TEST position.

4.5.5.8 Measure and record the attenuation of all harmonic and spurious R.F. power for all positions of the FREQ SELECT switch per para. 3.6.9.5.

4.5.5.9 Place the Control Assembly in the (Pulse) condition per para. 3.6.5.3.

4.5.5.10 The pulse spectrum for all positions of the FREQ SELECT switch shall be as defined in Figure 1 of para. 3.6.9.6.

4.5.5.11 Remove the spectrum analyzer and connect a frequency counter in its place.

4.5.5.12 Place the Control Assembly in the (CW) condition per para. 3.6.5.3.

4.5.5.13 Measure and record the self test output frequency for all positions of the FREQ SELECT switch per para. 3.6.9.4 (Table V).

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4.5.5.14 Remove the frequency counter and replace the 50 OHM termination on the CAL OUT port.

4.5.5.15 Place the Control Assembly in the (Pulse) condition per para. 3.6.5.3

4.5.5.16 Place the FUNCTION SELECT switch in the R.F. INPUT position.

4.5.5.17 Adjust the R.F. LEVEL attenuator such that the Digital Display indicates +1.000 ± 0.020 for all positions of the FREQ SELECT switch.

4.5.5.18 Place the Control Assembly in the (CW) condition per para. 3.6.5.3.

4.5.5.19 Set the R.F. LEVEL attenuator to 0 (zero).

4.5.5.20 Measure and record the Digital Display (DVM) indication for all positions of the FREQ SELECT switch per para. 3.6.9.8.

4.5.5.21 Place the MODE SELECT switch in the RADAR position.

4.5.5.22 Measure and record the Digital Display (DVM) indication for all positions of the FREQ SELECT switch per para. 3.6.9.8.1.

4.5.6. Cal Input

4.5.6.1 Perform initial condtions per para. 4.5.1.

4.5.6.2 Remove the 50 OHM termination from the CAL IN port and the CAL OUT port.

4.5.6.3 Inject a low level (approximately -5 dbm) signal into the CAL IN port at a frequency of 9,205 and 9,245 MHz ± 1 MHz.

4.5.6.4 Measure and record the insertion loss per para. 3.6.10.1.

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4.5.6.5 Replace the 50 OHM terminations.

4.5.7 R.F. Monitor

4.5.7.1 Perform initial conditions per para. 4.5.1.

4.5.7.2 Place the FUNCTION SELECT switch in the R.F. MONITOR position.

4.5.7.3 Place the Control Assembly in the "Pulse" condition per para. 3.6.5.3.

4.5.7.4 Place the RANGE SELECT switch in the R1 position.

4.5.7.5 Place the FREQ SELECT switch in the 9.205 position.

4.5.7.6 Adjust the R.F. LEVEL attenuator until the Digital Display (DVM) indicates +2.500 ± 0.050 per para. 3.6.11.4 Record R.F. LEVEL reading.

4.5.7.7 Place the FREQ SELECT switch in the remaining five positions and repeat para. 4.5.7.6.

4.5.7.8 Place the RANGE SELECT switch in the remaining three positions and repeat para. 4.5.7.5, 4.5.7.6, and 4.5.7.7.

4.5.7.9 Place the MODE SELECT switch in the RADAR position.

4.5.7.10 Place the R.F. LEVEL attenuator to 0 (zero).

4.5.7.11 Place the RANGE SELECT switch in the R3 position.

4.5.7.12 Measure and record the Digital Display (DVM) for all positions of the FREQ SELECT switch per para. 3.6.11.5.

4.5.7.13 Place the MODE SELECT switch in the SELF TEST position.

4.5.7.14 Adjust the R.F. LEVEL attenuator until the Digital Display (DVM) indicates +2.500 ± 0.020 per para. 3.6.11.6.

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4.5.7.15 Without further adjustment the Digital Display (DVM) shall indicate per para. 3.6.11.6.

4.5.7.16 Repeat para. 4.5.7.14 and 4.5.7.15 per para. 3.6.11.6.1 and Table VI.

4.5.7.17 Place the Control Assembly in the (CW) condition per para. 3.6.5.3.

4.5.7.18 Place the R.F. LEVEL attenuator at 0 (zero).

4.5.7.19 Measure and record the Digital Display (DVM) indication per para. 3.6.11.7.

4.5.8 Video Output

4.5.8.1 Perform initial conditions per para. 4.5.1.

4.5.8.2 Place the Control Assembly in the (Pulse) condition per para. 3.6.5.3.

4.5.8.3 Place the FUNCTION SELECT switch in the R.F. MONITOR position.

4.5.8.4 Remove the 50 OHM termination from the VIEDO OUT port and connect an oscilloscope in its place.

4.5.8.5 Adjust the R.F. LEVEL attenuator per para. 3.6.12.2.

4.5.8.6 Measure and record the video output per para. 3.6.12.3 for all positions of the FREQ SELECT switch.

4.5.8.7 Place the MODE SELECT switch in the RADAR position.

4.5.8.8 Place the R.F. LEVEL attenuator to 0 (zero).

4.5.8.9 Measure and record the video output per para. 3.6.12.4 for all positions of the FREQ SELECT switch.

4.5.8.10 Remove the oscilloscope and reconnect the 50 OHM termination to the VIDEO OUT port.

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4.5.9 Data Signals

4.5.9.1 Perform the initial conditions per para. 4.5.1.

4.5.9.2 Connect DATA INPUT cable W4 to connector J1 on the Control Assembly. Connect power supply to other end of W4.

4.5.9.3 With the FUNCTION SELECT switch in the positions delineated in Table VII, and applied voltages delineated in Table VII, measure and record the Digital Display (DVM) per para. 3.6.13.2

4.5.10 Signal Channel

4.5.10.1 Perform the initial conditions per para. 4.5.1.

4.5.10.2 Connect SIGNAL INPUT cable W2 to Control Assembly connector J3. Connect signal generator to other end of W2.

4.5.10.3 Apply signals per para. 3.6.14.2.

4.5.10.4 Measure and record the signal channel meter reading per para. 3.6.14.2.

4.5.11 Lock On Signal

4.5.11.1 Perform the initial conditions per para. 4.5.1.

4.5.11.2 Connect DATA INPUT cable W4 to Control Assembly connector J1. Connect power supply to other end of cable W4.

4.5.11.3 Apply signals per para. 3.6.15.2.

4.5.11.4 Record the lock on indicator condition per Table IX.

4.5.12 Elapse Time Indicator

4.5.12.1 This test can be performed during any three consecutive hours of running time.

4.5.12.2 Connect POWER SUPPLY cable W1 to Control Assembly connector J2.

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4.5.12.3 Record indication on elapsed time indicator at the conclusion of three consecutive hours of operation per para. 3.6.16.1.

4.5.13 Range and Phase Insertion

4.5.13.1 Perform the initial conditions per para. 4.5.1.

4.5.13.2 Place the Control Assembly in the (Pulse) condition per para 3.6.5.3.

4.5.13.3 Place the FUNCTION SELECT switch in the R.F. MONITOR position.

4.5.13.4 Place the RANGE SELECT switch in the R3 position.

4.5.13.5 Place the PHASE CONTROL knob in the approximate midrange position.

4.5.13.6 Adjust the R.F. LEVEL attenuator such that the Digital Display (DVM) indicates +2.500 $\pm$ 0.020.

4.5.13.7 Measure and record the Digital Display (DVM) indications for RANGE SELECT positions R1, R2, R3, and R4 without further adjustment of the R.F. LEVEL attenuator.

4.5.13.8 Repeat para. 4.5.13.4, 4.5.13.5, and 4.5.13.6.

4.5.13.9 Vary the PHASE CONTROL knob from one extreme (CCW) position to the other extreme (CW) position.

4.5.13.10 Record the Digital Display (DVM) indications at each extreme position per para. 3.6.16.4.

4.5.13.11 Repeat para. 4.5.13.4 thru 4.5.13.10 for all positions of the FREQ SELECT switch.

4.5.14 Clutter Signal Characteristics

4.5.14.1 Perform the initial conditions per para. 4.5.1.

4.5.14.1.1 Perform per para. 3.6.18.1 and 3.6.18.2 as required.

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4.5.14.2 Connect a spectrum analyzer and waveguide variable attenuator to the open end of cable W5.

4.5.14.3 Set the CLUTTER LEVEL dial to 5 (five).

4.5.14.4 Set the CLUTTER BALANCE attenuator to a minimum (completely CCW).

4.5.14.5 Measure and record the clutter signal per para. 3.6.18.6 for all positions of the FREQ SELECT switch.

4.5.14.6 Set the CLUTTER LEVEL dial to 20.

4.5.14.7 Place the CLUTTER switch to ON position.

4.5.14.8 Adjust the CLUTTER BALANCE attenuator per para. 3.6.18.7.

4.5.14.9 Measure and record the clutter signal per para. 3.6.18.7 for all positions of the FREQ SELECT switch.

4.5.14.10 Measure and record the harmonically related clutter power per para. 3.6.18.8 for all positions of the FREQ SELECT switch.

4.5.14.11 Measure and record the spurious related clutter power per para. 3.6.18.9 for all positions of the FREQ SELECT switch.

4.5.14.12 Set the R.F. LEVEL dial to 15.

4.5.14..13 Set the FREQ SELECT switch to 9.205 position.

4.5.14.14 Set the CLUTTER LEVEL dial to 5.

4.5.14.15 Measure and record the clutter signal level versus the CLUTTER LEVEL dial in dial increments of 10 until the CLUTTER LEVEL dial indicates 40 per para. 3.6.18.10 and Table X.

4.5.14.16 Repeat para. 4.5.14.14 and 4.5.14.15 with the FREQ SELECT switch in the 9.245 position.

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4.5.14.17 Remove the R.F. spectrum analyzer and waveguide variable attenuator from open end of cable W5 and connect a frequency counter.

4.5.14.18 Measure and record the clutter signal frequency for all positions of the FREQ SELECT switch per para. 3.6.18.11.

4.5.14.19 Connect an R.F. power meter to cable W5.

4.5.14.20 Place the FREQ SELECT switch in the 9.229 position.

4.5.14.21 Set the CLUTTER LEVEL dial at 20 and FUNCTION SELECT switch to R.F. MONITOR. Adjust R.F. LEVEL control for a reading of 2.500 $\pm$ 0.020 on the Digital Display (DVM).

4.5.14.22 Adjust the CLUTTER BALANCE attenuator such that the clutter signal (CW) is -20.0 $\pm$ 0.1 dbm.

4.5.14.23 Remove the R.F. power meter and connect cable W5 to the Control Assembly SELF TEST port.

4.5.14.24 Place the FUNCTION SELECT switch in the SELF TEST position.

4.5.14.25 Place the Control Assembly in the (Pulse) condition per para. 3.6.5.3.

4.5.14.26 Measure and record the Digital Display (DVM) per para. 3.6.18.12.

4.5.14.27 Repeat para. 4.5.14.19 thru 4.5.14.26 for FREQ SELECT switch positions 9.205 and 9.245.

4.5.14.28 Connect an R.F. power meter to cable W5.

4.5.14.29 Set the CLUTTER LEVEL dial to 5.

4.5.14.30 Measure and record the change in clutter signal (CW) when the R.F. LEVEL dial is increased a 5 increment from its internal R.F. power calibration reference dial setting per para. 3.6.18.13 for all positions of the FREQ SELECT switch.

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4.5.14.31 Remove the R.F. power meter and connect a 20 db coaxial attenuator and analyzer to cable W5.

4.5.14.32 Place the Control Assembly in the (Pulse) condition per para. 3.6.5.3.

4.5.14.33 Measure and record the pulse spectrum per para. 3.6.18.14 for all positions of the FREQ SELECT switch.

4.5.15 Target Signal Characteristics

4.5.15.1 Perform the initial conditions per para. 4.5.1

4.5.15.1.1 Perform per para. 3.6.19.1 and 3.6.19.2 as required.

4.5.15.2 Connect a spectrum analyzer and waveguide variable attenuator to the open end of cable W5.

4.5.15.3 Set the TARGET LEVEL dial to 27.

4.5.15.4 Set the TARGET BALANCE attenuator to a minimum (completely CCW).

4.5.15.5 Measure and record the target signal per para. 3.6.19.6 for all positions of the FREQ SELECT switch.

4.5.15.6 Place the TARGET switch to ON position.

4.5.15.7 Adjust the TARGET BALANCE attenuator per para. 3.6.19.7.

4.5.15.8 Measure and record the target signal per para. 3.6.19.7 for all positions of the FREQ SELECT switch.

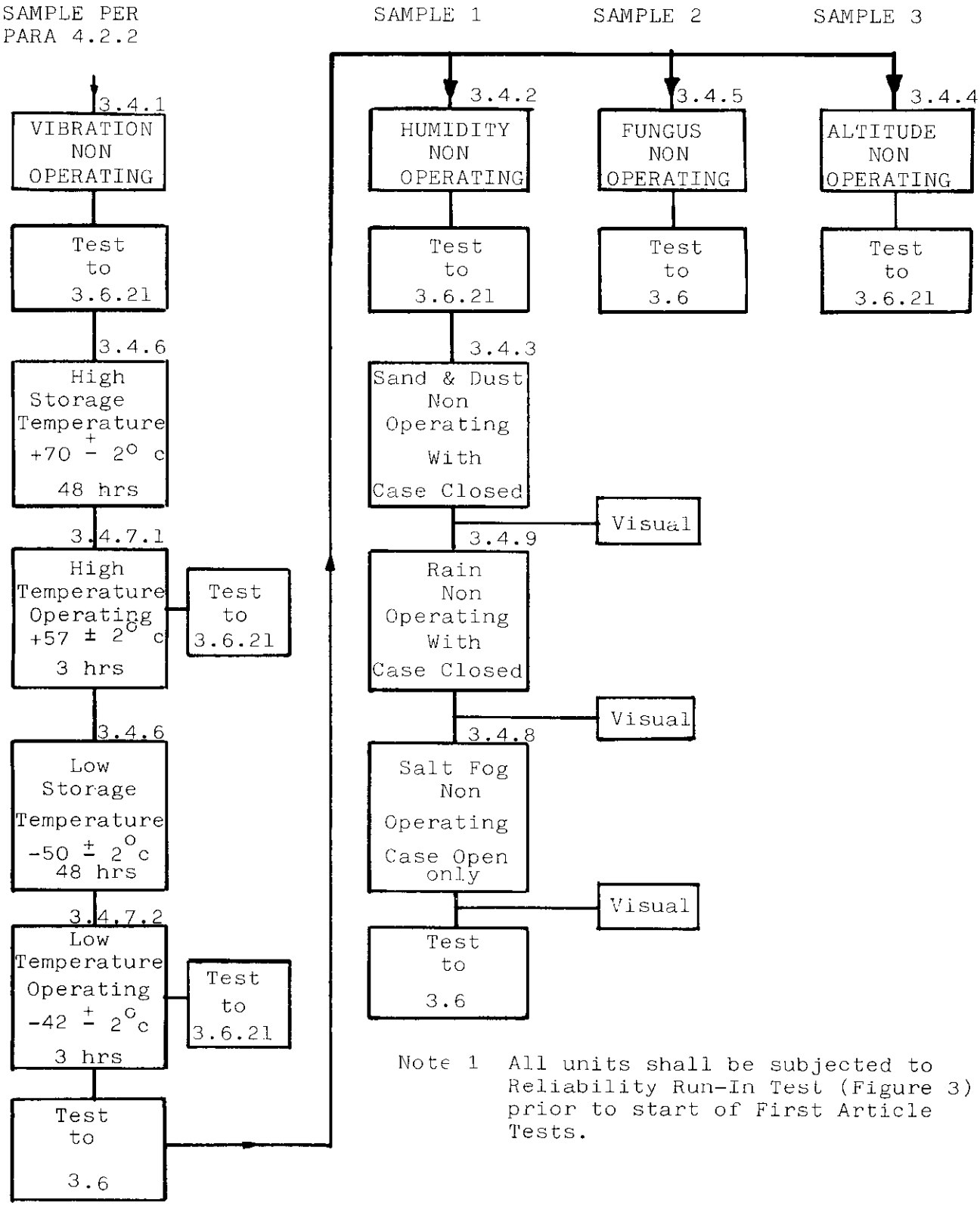
4.5.15.9 Place the CLUTTER switch to the ON position and adjust the clutter signal (within 10 db) of a target signal reference.

4.5.15.10 Measure and record the frequency difference between the clutter signal and the target signal per para. 3.6.19.8 for all positions of the FREQ SELECT switch.

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FIGURE 4

AN/VPM-2 FIRST ARTICLE & CONTROL TEST PLAN



Note 1 All units shall be subjected to Reliability Run-In Test (Figure 3) prior to start of First Article Tests.

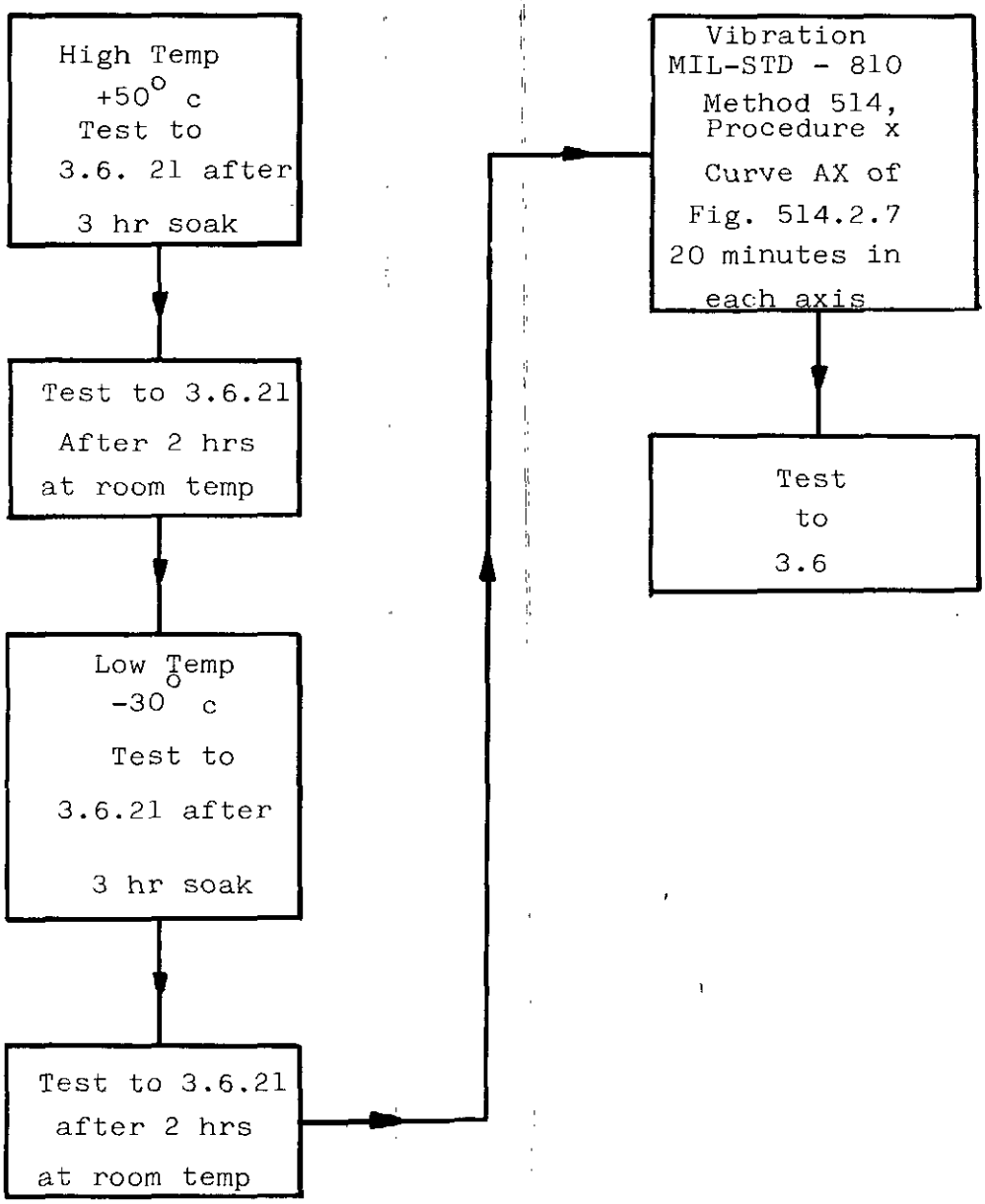
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FIGURE-5

AN/VPM-2 POWER SUPPLY

RELIABILITY RUN-IN (TEMPERATURE & VIBRATION)

TEST PLAN



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4.5.15.11 Place the CLUTTER switch to the OFF position, and the TARGET LEVEL to 27.

4.5.15.12 Measure and record the harmonically and spurious related sidebands; Control Assembly in (CW) condition; per para. 3.6.19.9, for all positions of the FREQ SELECT switch.

4.5.15.13 Set the R.F. LEVEL dial to 15.

4.5.15.14 Set the FREQ SELECT switch to the 9.205 position.

4.5.15.15 Set the TARGET LEVEL dial to 27.

4.5.15.16 Measure and record the target signal level versus the TARGET LEVEL dial increments of 10 until the TARGET LEVEL dial indicates 87 per para. 3.6.19.11 and Table X.

4.5.15.17 Repeat para. 4.5.15.13 thru 4.5.15.16 with the FREQ SELECT switch in the 9.245 position.

4.5.15.18 Remove the spectrum analyzer and connect a diode detector, R.F. amplifier and oscilloscope.

4.5.15.19 Measure and record the target signal output with the TARGET FREEZE and TARGET switch combinations per para. 3.6.19.12 and Table XI.

4.5.15.20 Remove the diodedetector, R.F. amplifier and oscilloscope from cable W5.

4.5.15.21 Connect cable W5 to the Control Assembly SELF TEST port.

4.5.15.22 Place the CLUTTER switch in the ON position, the TARGET switch in the OFF position and the FREQ SELECT switch to 9.229.

4.5.15.23 Place the Control Assembly in the (Pulse) condition per para. 3.6.5.3.

4.5.15.24 Adjust the CLUTTER BALANCE attenuator until the Digital Display (DVM) indicates the value per para. 3.6.18.12, (Self Test Calibration value).

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4.5.15.25 Place the CLUTTER LEVEL dial at 25 and the TARGET LEVEL dial at 27. Turn the CLUTTER switch ON and the TARGET switch OFF.

4.5.15.26 Record the Digital Display (DVM) indication.

4.5.15.27 Place the CLUTTER switch in the OFF position and the TARGET switch in the ON position.

4.5.15.28 Adjust the TARGET BALANCE attenuator until the Digital Display (DVM) indicates the value recorded in para. 4.5.15.26. The Digital Display shall indicate per para. 3.6.19.13.

4.5.15.29 Disconnect cable W5 from the SELF TEST port and connect a spectrum analyzer to the open end of cable W5.

4.5.15.30 Place the Control Assembly in the (CW) condition per para. 3.6.5.3.

4.5.15.31 The target signal level and the clutter signal level shall be as delineated in para. 3.6.19.12.1 and 3.6.19.12.2.

4.5.15.32 Set the TARGET LEVEL dial at 27.

4.5.15.33 Measure and record the change in target signal (CW) when the R.F. LEVEL dial is increased a 5 increment from its internal R.F. power calibration reference dial setting per para. 3.6.19.1.

4.5.15.34 Place the Control Assembly in the (Pulse) condition per para. 3.6.5.3.

4.5.15.35 Measure and record the pulse spectrum per para. 3.6.19.14 for all positions of the FREQ SELECT switch.

4.5.16 Calibration Data

4.5.16.1 Perform the initial conditions per para. 4.5.1.

4.5.16.2 Place the FUNCTION SELECT switch in the R.F. INPUT position.

4.5.16.2.1 Connect cable W5 to the RADAR port.

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4.5.16.3 Inject an R.F. peak power as delineated in para. 3.6.20.1 having a frequency of 9,229 ± 1 MHz into the open end of cable W5.

4.5.16.4 Adjust the R.F. LEVEL attenuator until the Digital Display (DVM) indicates +2.500 ± 0.020.

4.5.16.5 Determine the R.F. INPUT calibration number by subtracting the R.F. LEVEL attenuator dial setting from the R.F. peak power input. The value shall be within the requirements of para. 3.6.20.1.1.

4.5.16.5.1 Record R.F. INPUT calibration data per para. 3.6.20.1.2.

4.5.16.6 Repeat para. 4.5.16.3 and 4.5.16.4 for R.F. peak power inputs having frequencies of 9,205 ± 1 MHz and 9,245 ± 1 MHz.

4.5.16.6.1 Record R.F. LEVEL attenuator dial settings.

4.5.16.6.2 The R.F. INPUT calibration number determined in para. 4.5.16.5 when added to the R.F. LEVEL attenuator dial settings of para. 4.5.16.6.1 shall measure the R.F. input peak power per para. 3.6.20.1.

4.5.16.7 Place the FUNCTION SELECT switch in the R.F. SELF TEST position.

4.5.16.8 Inject into the Control Assembly SELF TEST port an R.F. peak power having characteristics as delineated in para. 3.6.20.2 at a frequency of 9,229 ± 1 MHz.

4.5.16.9 The Digital Display (DVM) should indicate a value between +1.000 and +2.500.

4.5.16.10 Record the value (para. 4.5.16.9) as the R.F. SELF TEST calibration data per para. 3.6.20.2.2.

4.5.16.11 Place the MODE SELECT switch in the SELF TEST position and the FREQ SELECT switch in the 9.229 position.

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4.5.16.12 Place the Control Assembly in the (Pulse) condition per para. 3.6.5.3.

4.5.16.13 Place the RANGE SELECT switch in the R1 position.

4.5.16.14 Connect a Detector and an R.F. amplifier to the CAL OUT port and connect video end of Detector through a coaxial T to CH A START on counter. Connect the other arm of the coaxial T to CH A of Oscilloscope. Ensure Oscilloscope is DC COUPLED.

4.5.16.15 Connect another Detector and R.F. amplifier to open end of cable W5. (Use W/G to Coax Adapter.) Connect video end of Detector through a coaxial T to CH B STOP on counter. Connect the other arm of the coaxial T to CH B of Oscilloscope. Ensure Oscilloscope is DC COUPLED.

4.5.16.16 Set CLUTTER switch to ON position and TARGET switch to OFF position.

4.5.16.17 Adjust Oscilloscope until pulse display is positioned with the 50% point of the pulse amplitude on the center graticule of the vertical display.

4.5.16.18 Disconnect Detector lead from Oscilloscope CH A.

4.5.16.19 Connect CH A LEVEL (located on rear of counter) to Oscilloscope CH A. (Do not change vertical sensitivity of Oscilloscope).

4.5.16.20 Adjust LEVEL CONTROL on counter for level display on the center graticule of the vertical display of the oscilloscope.

4.5.16.21 Repeat para. 4.5.16.18 thru 4.5.16.21 for CH B. (TARGET switch OFF position, CLUTTER switch ON position).

NOTE

Pulse should have sharp rise time. To sharpen leading edge of the pulse, add coaxial T's to both CH A and B of counter or Oscilloscope and add resistive loading.

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4.5.16.22 Set Counter controls for time interval A to B. Set gate time for 1 nanosecond resolution. Read delay on counter display.

4.5.16.23 Convert the delay (para. 4.5.16.23) from nanoseconds to meters per para. 3.6.20.3.2.

4.5.16.24 Record the range delay in meters as the RANGE calibration data per 3.6.20.3.4.

4.5.16.25 Repeat para. 4.5.16.14 thru 4.5.16.5 for RANGE SELECT positions R2, R3, and R4.

4.5.16.26 The range delay (R1, R2, R3, R4) for FREQ SELECT positions 9,205 ± 1 MHz and 9,245 ± 1 MHz shall be measured (para. 4.5.16.14 thru 4.5.16.24) and shall be as delineated in para. 3.6.20.3.1.

4.5.17 Radar Test

4.5.17.1 Connect the power cables (W1, W3) and signal cables (W2, W4) as shown in Figure 3. (Signal cables are not necessary for SELF TEST procedures and are connected at this time only for convenience.) Do not connect the flexible wave guide cable W5.

4.5.17.2 With equipment properly interconnected, set CONTROL ASSEMBLY controls as delineated in Table XV.

TABLE XV

Control	Position
POWER	OFF
FUNCTION SELECT	RF MON
VOLTAGE SELECT	+24
R.F. LEVEL	20 db
FREQ SELECT	9.229
MODE SELECT	SELF TEST

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TABLE XV, Continued

PHASE CONTROL	MID RANGE
CLUTTER	OFF
CLUTTER LEVEL	20 db
RANGE SELECT	R3
TARGET LEVEL	20 db
TARGET FREEZE	OFF
TARGET	OFF

4.5.17.3 Set the Radar Set to the Standby mode (TM 9-1285-200-34/3).

4.5.17.4 Set POWER switch to ON position. Ensure that Digital Display (DVM) is illuminated.

4.5.17.5 Connect one end of flexible wave guide W5 to RADAR port (J5) and other end to SELF TEST port (J6) on front panel of the Control Assembly.

Note

Ensure pin keying on flexible waveguide cable W5 flanges is correct.

4.5.17.6 Set CLUTTER switch to ON position. Set CLUTTER LEVEL dial to 20.

4.5.17.7 Adjust R.F. LEVEL control to obtain indication of +2.500 $\pm$ 0.025 on Digital Display (DVM).

NOTE

Paragraph 4.5.17.7 ensures that a calibrated power level is injected into the target and clutter channels. Therefore, monitor and maintain the adjustment delineated in paragraph 4.5.17.7 throughout the remaining procedures unless otherwise indicated.

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4.5.17.8 Set FUNCTION SELECT switch to R.F. SELF TEST position.

4.5.17.9 Adjust CLUTTER BALANCE for an indication on Digital Display (DVM) as delineated on System Tester Calibration decal.

IMPORTANT: MAINTAIN R.F. MONITOR LEVEL PARAGRAPH 4.5.17.7 DURING ACCOMPLISHMENT OF PARA. 4.5.17.9.

4.5.17.10 Set CLUTTER LEVEL control to 25. Set CLUTTER switch to OFF position.

4.5.17.11 Set TARGET switch to ON position. Set TARGET LEVEL to 27.

4.5.17.12 Adjust R.F. LEVEL control to obtain an indication of +0.500 vdc to ± 1.000 vdc on the Digital Display (DVM).

4.5.17.13 Set TARGET switch to OFF position. Set CLUTTER switch to ON position.

4.5.17.14 Observe and note Digital Display (DVM) indication.

4.5.17.15 Set CLUTTER switch to OFF position and TARGET switch to ON position.

4.5.17.16 Adjust TARGET BALANCE control for same Digital Display (DVM) indication as in para. 4.5.17.14

4.5.17.17 Alternately repeat para. 4.5.17.13 thru 4.5.17.16 until Digital Display (DVM) indications in para. 4.5.17.14 and 4.5.17.16 are within ± 0.025 vdc.

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4.5.17.18 Disconnect flexible wave guide cable W5 from SELF TEST port (J6) of the Control Assembly.

4.5.17.19 Set Control Assembly controls as delineated in Table XVI.

TABLE XVI

Control	Position
FUNCTION SELECT	VOLTAGE
VOLTAGE SELECT	+24
RF LEVEL	20 db
FREQ SELECT	RADAR CRYSTAL FREQUENCY +60 MHz
MODE SELECT	RADAR
PHASE CONTROL	MID RANGE
CLUTTER	OFF
CLUTTER LEVEL	70 db
RANGE SELECT	R1
TARGET LEVEL	70 db
TARGET FREEZE	OFF
TARGET	OFF
POWER	ON

4.5.17.20 Connect flexible waveguide cable W5 to Radar Set AN/VPS-2 R.F. OUTPUT/INPUT port. (When mounting do not cross-guide waveguide.)

4.5.17.21 Observe Digital Display (DVM) for proper Power Input voltage (+24.00 $\pm$ 6.000 vdc).

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4.5.17.22 Set VOLTAGE SELECT switch to other positions in sequence (+5, +15, -15, +28) and observe Digital Display (DVM) for correct output voltage indications (+5.00 ± 0.200 vdc, +15.00 ± 0.300 vdc, -15.00 ± 0.300 vdc, +28.00 ± 0.600 vdc).

4.5.17.23 Energize Radar Set to RADIATE POWER conditions by setting SERV RAD CONT switch S6 inside top cover of Transmitter Receiver Unit 2 to ON.

4.5.17.24 Ensure that LOCK ON indicator is not illuminated and SIGNAL CHANNEL meter indicaaes less than 1.0 vdc.

4.5.17.25 Set FUNCTION SELECT switch to R.F. MONITOR position.

4.5.17.26 Adjust R.F. LEVEL control for indication of +2.500 ± 0.025 on Digital Display (DVM).

NOTE

Paragraphs 4.5.17.25 and 4.5.17.26 ensure that a calibrated signal is injected into the target and clutter channels. Therefore, they should be monitored and maintained throughout the following procedures, unless otherwise indicated.

4.5.17.27 R.F. Transmitted Peak Power Measurement Procedure

4.5.17.27.1 Set FUNCTION SELECT switch to R.F. INPUT position.

4.5.17.27.2 Adjust R.F. LEVEL Control for an indication of +2.500 ± .050 on Digital Display (DVM).

4.5.17.27.3 Determine the Radar Set R.F. Peak Transmitted Power by the following calculation:

R.F. LEVEL setting (paragraph 4.5.17.27.2) + R.F. INPUT on System Tester Calibration decal = Radar Set Peak Transmitter Power in dbm.

4.5.17.28 Minimum Discernible Signal (MDS) Procedure.

4.5.17.28.1 Set TARGET switch to ON position.

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4.5.17.28.2 With FUNCTION SELECT switch in R.F. MONITOR position, adjust the R.F. LEVEL Control and maintain on Digital Display (DVM) an indication of 2.500 ± 0.025 vdc for the remainder of this procedure.

4.5.17.28.3 Decrease (lower number on dial) TARGET LEVEL control until LOCK ON indicator illuminates. Note TARGET LEVEL dial number.

4.5.17.28.4 Increase (higher number on dial) TARGET LEVEL setting by 10.0 db. LOCK ON light should not be illuminated after an approximate 3 second delay.

4.5.17.28.5 Repeat para. 4.5.17.28.3 and 4.5.17.28.4 until repeatability occurs in para. 4.5.17.28.3 within 1.0 db.

4.5.17.28.6 Record TARGET LEVEL dial setting. This is the MDS of the Radar Set.

4.5.17.29 Range Accuracy Measurement Procedure

4.5.17.29.1 Set TARGET switch to ON position.

4.5.17.29.2 Adjust TARGET LEVEL control until LOCK ON indicator illuminates.

4.5.17.29.3 Decrease (lower number on dial) TARGET LEVEL control setting by 10.

4.5.17.29.4 Set FUNCTION SELECT switch to RANGE position.

4.5.17.29.5 Read and record Range voltage indicated on Digital Display (DVM).

4.5.17.29.6 Set RANGE SELECT switch to R2, R3, and R4, positions and repeat para. 4.5.17.29.5.

NOTE: Allow 3 seconds for the Radar to re-acquire the target signal each time the RANGE SELECT switch position is changed.

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4.5.17.29.7 Set FUNCTION SELECT switch to Range REF position.

4.5.17.29.8 Read and record Range Reference voltage indicated on Digital Display (DVM).

4.5.17.29.9 The RANGE REF acceptability criteria is +20.00 ± 0.200 vdc. If the RANGE REF measurement does not meet this criteria the Radar Set must be adjusted or repaired.

4.5.17.29.10 Determine Range of the Radar Set by the following calculation:

$$\text{Calculated Range (meters)} = \frac{\text{Range voltage (para. 4.5.17.29.5) x 5000}}{\text{Range Ref. voltage (para 4.5.17.29.8)}}$$

4.5.17.30 Sub-Clutter Visibility (SCV) Measurement Procedure

4.5.17.30.1 Connect TPS short on J6 of Radar Receiver Unit 3.

4.5.17.30.2 Set RANGE SELECT to R4.

4.5.17.30.3 Set TARGET switch to ON position.

4.5.17.30.4 Adjust TARGET LEVEL control until LOCK ON indicator is illuminated.

4.5.17.30.5 Decrease (lower number on dial) TARGET LEVEL control setting by 5 db.

4.5.17.30.6 Set TARGET FREEZE switch to ON position. (LOCK ON indicator should stay illuminated.)

4.5.17.30.7 Set CLUTTER switch to ON position.

4.5.17.30.8 Adjust CLUTTER LEVEL control until SIGNAL CHANNEL meter indicates 1 to 2 volts.

4.5.17.30.9 Adjust PHASE CONTROL for maximum signal indication on SIGNAL CHANNEL meter.

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4.5.17.30.10 Set TARGET, TARGET FREEZE, and CLUTTER switches to OFF position.

4.5.17.30.11 Set CLUTTER LEVEL control to 70.

4.5.17.30.12 Set CLUTTER switch to ON position.

4.5.17.30.13 With FUNCTION SELECT switch in R.F. MONITOR position, adjust with R.F. LEVEL control and maintain on Digital Display (DVM) an indication of +2.500 ± 0.025 vdc for the remainder of this procedure.

4.5.17.30.14 Adjust CLUTTER LEVEL control until LOCK ON indicator is illuminated. Note CLUTTER LEVEL dial number.

4.5.17.30.15 Increase (higher number on dial) CLUTTER LEVEL control setting by 10.0.

4.5.17.30.16 LOCK ON indicator should not be illuminated after an approximate 3 second delay.

4.5.17.30.17 Repeat para. 4.5.17.30.14 thru 4.5.17.30.16 until repeatability occurs in para. 4.5.17.30.14 within 1.0 db.

4.5.17.30.18 Record CLUTTER LEVEL dial reading (para. 4.5.17.30.14). This is the CLUTTER POWER.

4.5.17.30.19 Determine SCV of the Radar Set by the following calculation:

SCV = MDS (Para. 4.5.17.28.6) - Clutter Power (para. 4.5.17.30.18).

4.5.17.31 AGC Linearity Measurement of Radar

4.5.17.31.1 Disconnect TPS short from J6 of Radar Receiver Unit 3.

4.5.17.31.2 With FUNCTION SELECT switch in R.F. MONITOR position, adjust R.F. LEVEL control and maintain on Digital Display (DVM) an indication of +2.500 ± 0.025 vdc for the remainder of this procedure.

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4.5.17.31.3 Set CLUTTER switch to ON position. Decrease CLUTTER LEVEL dial setting slowly to 5 while observing LOCK ON indicator. LOCK ON indicator should not illuminate.

4.5.17.31.4 Set CLUTTER switch to OFF position.

4.5.17.31.5 Set TARGET switch to ON position.

4.5.17.31.6 Adjust TARGET LEVEL to MDS (para. 4.5.17.28.6) LOCK ON indicator should illuminate.

4.5.17.31.7 Set CLUTTER switch to ON position, LOCK ON indicator should not illuminate.

4.5.17.31.8 Increase (higher number on dial) CLUTTER LEVEL control setting until LOCK ON indicator is illuminated. Note CLUTTER LEVEL dial reading.

4.5.17.31.9 Decrease (lower number on dial) CLUTTER LEVEL setting by 10.0. LOCK ON indicator should not be illuminated after an approximate 3 second delay.

4.5.17.31.10 Repeat para. 4.5.17.31.8 and 4.5.17.31.9 until repeatability occurs in para. 4.5.17.31.8 within 1.0 db.

4.5.17.31.11 Record CLUTTER LEVEL dial reading. This is the CLUTTER POWER.

4.5.17.31.12 Determine the AGC Threshold by the following calculation:

AGC Threshold = MDS (para 4.5.17.28.6) - Clutter Power (para. 4.5.17.31.11)

4.5.17.31.13 Decrease (lower number on dial) CLUTTER LEVEL control setting by 10 and record dial reading. The LOCK ON indicator should not be illuminated after an approximate 3 second delay.

4.5.17.31.14 Decrease TARGET LEVEL control setting until LOCK ON indicator illuminates. Record TARGET LEVEL dial reading.

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4.5.17.31.15 Repeat steps para. 4.5.17.31.13 and 4.5.17.31.14 in 10 db increments. The last increment may be less than 10. (Record each increment.)

4.5.17.31.16 Determine AGC Linearity by the Following calculation: (para. 4.5.17.31.13) - (4.5.17.31.14) = Incremental AGC Linearity.

4.5.17.32 Ancillary Measurements of Range Rate Reference (+) and Range Rate Reference (-).

4.5.17.32.1 Set FUNCTION SELECT switch to RANGE RATE REF (+) position.

4.5.17.32.2 Record reading indicated on Digital Display (DVM).

4.5.17.32.3 Set FUNCTION SELECT switch to RANGE RATE REF (-) position.

4.5.17.32.4 Record reading indicated on Digital Display (DVM).

4.5.17.32.5 This completes the RADAR TEST procedures.

4.6 Reliability Run-IN. - The following run-in tests shall be performed on a 100% basis.

4.6.1 Temperature Run-In. The Control Assembly shall be tested in accordance with the requirements of Figure 5.

4.6.2 Vibration Run-In. The Control Assembly shall be tested in accordance with the requirements of Figure 5.

4.7 Packaging, packing, and marking. Quality assurance provisions for packaging, packing, and marking shall be as specified in MIL-P-14232 and Section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet 11748205. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended Use. - The Control Assembly is used to provide R.F. and digital control functions for testing the AN/VPS-2 VULCAN RADAR SET.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of preservation, packaging, and packing.
c. Applicable stock number.
d. Applicable packaging data sheet (see 5.1).
e. First article sample requirements.

6.3 Drawings. - Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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MIL-C-48605 Control Assembly R.F. (UNIT) 11748205

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